Homework 7 Solutions

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Section 3.1

3.1-3

Given $X \sim U(0, 10)$.

(a) The pdf of X:

$$f(x) = \begin{cases} \frac{1}{10}, & 0 < x < 10 \\ 0, & \text{otherwise} \end{cases}$$

(b) $P(X \geq 8) = \int_8^{10} \frac{1}{10} dx = 0.2$

(c) $P(2 \leq X < 8) = \int_2^8 \frac{1}{10} dx = 0.6$

(d) $E(X) = \frac{a+b}{2} = 5$

(e) $Var(X) = \frac{(b-a)^2}{12} = \frac{25}{3}$

3.1-4

Given $M(t) = \frac{e^{5t} - e^{4t}}{t}$.
From the form, $X \sim U(4, 5)$.

(a) $E(X) = \frac{4+5}{2} = 4.5$

(b) $Var(X) = \frac{(5-4)^2}{12} = \frac{1}{12}$

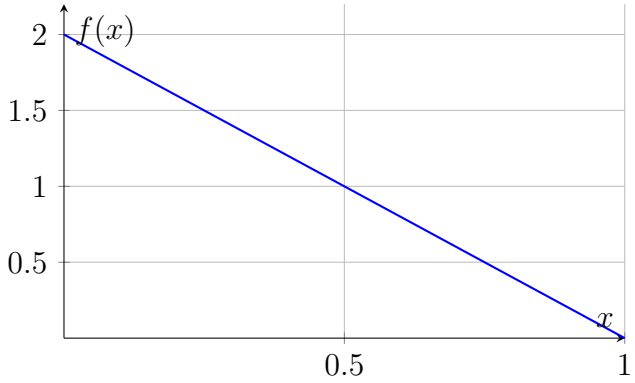
(c) $P(4.2 < X \leq 4.7) = 4.7 - 4.2 = 0.5$

3.1-9

Given $f(x) = 2(1 - x),\ 0 \leq x \leq 1$.

(a) The pdf is a straight line from $(0, 2)$ to $(1, 0)$.

PDF of $f(x) = 2(1-x)$



(b) $F(x) = 2x - x^2$ for $0 \leq x \leq 1$.

(c)

$$(i)\ P(0 \leq X \leq 1/2) = F(1/2) = \frac{3}{4}$$

$$(ii)\ P(1/4 \leq X \leq 3/4) = F(3/4) - F(1/4) = \frac{1}{2}$$

$$(iii)\ P(X = 3/4) = 0$$

$$(iv)\ P(X \geq 3/4) = 1 - F(3/4) = \frac{1}{16}$$

3.1-11

Given $g(y) = \frac{c}{y^3},\ y > 1$.

(a) Find c:

$$\int_1^\infty \frac{c}{y^3} dy = 1 \Rightarrow c = 2$$

(b) $E(Y) = \int_1^\infty y \cdot \frac{2}{y^3} dy = 2$

(c) $E(Y^2)$ diverges, so $Var(Y)$ is not finite.

3.1-16

Given $f(x) = \frac{x+1}{2},\ -1 < x < 1$.

$$F(x) = \frac{(x+1)^2}{4}$$

Solve $F(\pi_p) = p$:

$$\pi_p = -1 + 2\sqrt{p}$$

(a) $\pi_{0.64} = 0.6$

(b) $q_1 = \pi_{0.25} = 0$

(c) $\pi_{0.81} = 0.8$

Section 3.2

3.2-1

(a) $M(t) = \frac{1}{1-3t}$
$\Rightarrow$ Exponential with $\theta = 3$
$f(x) = \frac{1}{3}e^{-x/3}$, $E(X) = 3$, $Var(X) = 9$

(b) $M(t) = \frac{3}{3-t} = \frac{1}{1-t/3}$
$\Rightarrow$ Exponential with $\theta = \frac{1}{3}$
$f(x) = 3e^{-3x}$, $E(X) = \frac{1}{3}$, $Var(X) = \frac{1}{9}$

3.2-2

Rate $\lambda = \frac{2}{3}$ per minute, so $\theta = 1.5$.

(a) $f(x) = \frac{2}{3}e^{-2x/3},\ x > 0$

(b) $P(X > 2) = e^{-4/3}$

3.2-6

Average 3 flaws per 100 ft $\Rightarrow \lambda = 0.03$ flaws/ft.

(a) Expected flaws in first 40 ft: $\mu = 0.03(40) = 1.2$

$$P(0 \text{ flaws}) = e^{-1.2} \approx 0.301$$

(b) Assumptions:

- Flaws occur independently.
- Constant rate.
- Events occur one at a time.